# Project BattleBot

## Line-Following Track


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## Line-Following Track – Setup and Description

The robotic system is built on an Arduino Nano platform and designed to operate on a curved line-following track. The setup integrates multiple sensors and actuators to enable autonomous navigation, object interaction, and real-time decision-making.

## Hardware Components

The system consists of the following main components:

- 8 Analog Line Sensors (A0–A7)

  Used to detect the position of the line relative to the robot.
- Ultrasonic Sensor (TRIG: 7, ECHO: 8)

  Used for both start signal detection (flag) and obstacle detection.
- DC Motors (PWM control)
  - Left motor: pins 3 (back), 5 (forward)
  - Right motor: pins 11 (back), 6 (forward)

    Responsible for movement and steering.
- Servo Motor (Pin 9)

  Controls the gripper mechanism to grab and release objects.
- NeoPixel LEDs (Pin 12)

  Provide visual feedback for robot states (e.g., movement, turning, stopping).

## Inputs (Sensors → Robot)

| Name | Type | Pin | Description |
|---|---|---|---|
| Ultrasonic Trigger | Digital Output | 7 | Sends signal to measure distance |
| Ultrasonic Echo | Digital Input | 8 | Receives reflected signal |
| Line Sensor 0 | Analog Input | A0 | Left-most line detection |
| Line Sensor 1 | Analog Input | A1 | Line detection |

| | | | |
|---|---|---|---|
| Line Sensor 2 | Analog Input | A2 | Line detection |
| Line Sensor 3 | Analog Input | A3 | Line detection |
| Line Sensor 4 | Analog Input | A4 | Line detection |
| Line Sensor 5 | Analog Input | A5 | Line detection |
| Line Sensor 6 | Analog Input | A6 | Line detection |
| Line Sensor 7 | Analog Input | A7 | Right-most line detection |

### Logical Inputs (Software-Level)

| Name | Source | Purpose |
|---|---|---|
| Flag Detection | Ultrasonic | Detect start condition |
| Obstacle Detection | Ultrasonic | Trigger avoidance |
| Line Position | Line sensors | Control steering |
| End Detection | Line sensors | Detect finish (all black) |

## Outputs (Robot → Actions)

| Name | Type | Pin | Description |
|---|---|---|---|
| Motor Left Forward | PWM | 5 | Moves left wheel forward |
| Motor Left Backward | PWM | 3 | Moves left wheel backward |
| Motor Right Forward | PWM | 6 | Moves right wheel forward |
| Motor Right Backward | PWM | 11 | Moves right wheel backward |
| Servo Gripper | PWM | 9 | Opens/closes gripper |
| NeoPixel LED | Digital | 12 | Status lighting |

### Logical Outputs (Behavior)

| Name | Action |
|---|---|
| driveForward() | Move robot forward |
| pivotLeft() | Turn left |

| | |
|---|---|
| pivotRight() | Turn right |
| driveBackward() | Reverse |
| openGripper() | Release object |
| closeGripper() | Grab object |
| LED feedback | Status indication |

## System Description

The robot follows an **Input–Process–Output architecture**, where sensor data is continuously processed to control movement and behavior.

### 1. Input Layer

The robot collects data from:

- Line sensors → detect line position
- Ultrasonic sensor → detect flag and obstacles

### 2. Processing Layer

The core logic is implemented using a **finite state machine (FSM)**:

States include:

- WAIT_FOR_FLAG → start condition
- DRIVE, TURN, LINE_FOLLOW → navigation
- AVOID_* → obstacle avoidance
- STOPPED, BACKWARD, OPEN → end sequence

The system also uses:

- **Weighted sensor calculation** to determine position
- **Error-based correction** for steering
- **Timing control (millis)** for non-blocking actions

### 3. Output Layer

Based on decisions, the robot controls:

- Motors → movement and direction
- Servo → object handling
- LEDs → status indication

## Behavior Flow

The robot follows a structured sequence of actions:

1. **Start Detection**

   The robot waits until the flag is removed using ultrasonic sensing.

2. **Object Interaction**

   The robot opens its gripper, moves forward, and grabs an object.

3. **Navigation Phase**

   The robot:

   - Turns toward the track
   - Enters line-following mode
   - Continuously adjusts direction based on sensor input

4. **Obstacle Avoidance**

   If an obstacle is detected:

   - The robot executes a predefined avoidance path
   - Then returns to the line

5. **End Detection**

   When all sensors detect black:

   - The robot stops
   - Moves backward
   - Releases the object

### Behavior Flow Diagram

```
INPUT (Sensors)
        ↓
Read Distance + Line Sensors
        ↓
PROCESS (State Machine)
        ↓
Decision Making:
  - Follow line?
  - Avoid obstacle?
  - Grab/release?
        ↓
OUTPUT
  - Motors move
  - Servo acts
  - LEDs show status
```

## Line-Following Mechanism

The robot uses a weighted sensor system to determine its position:

- Each sensor has a predefined weight
- The system calculates an average position
- The error value represents deviation from the center

Motor speeds are adjusted using:

- Higher error → stronger correction
- Lower error → higher speed

This creates a smooth, proportional control behavior, suitable for curved tracks.

### Code Mapping (Design Connection)

The implementation directly reflects the system design:

- Input handling
  - analogRead(SENSOR_PINS[i]) → line detection
  - readDistance() → ultrasonic sensing
- Processing logic
  - switch(state) → state machine
  - lineFollow() → navigation logic
- Output control
  - analogWrite() → motor control
  - servoPulse → gripper control
  - setColor() → LED feedback

### Summary

The system is designed to operate on a line-following track, requiring continuous real-time adjustment. By combining multiple sensors, a state machine, and proportional control logic, the robot achieves stable navigation, obstacle handling, and task execution in a dynamic environment.